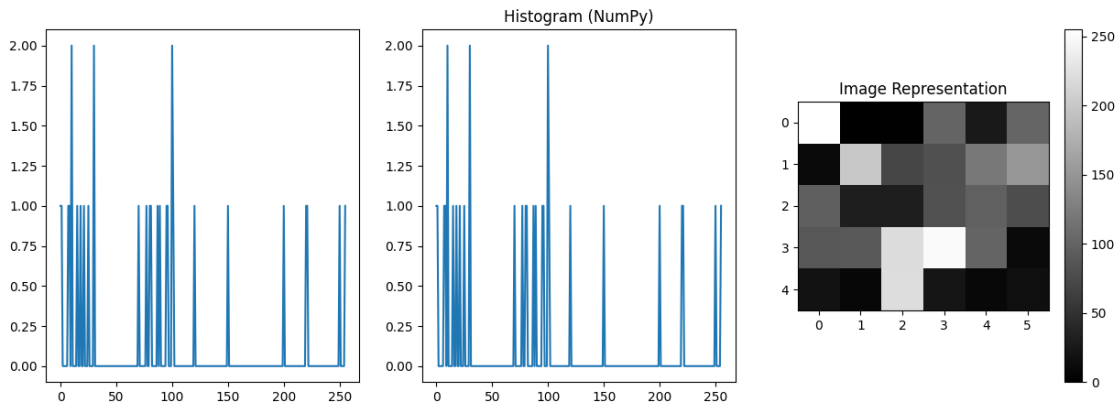
1

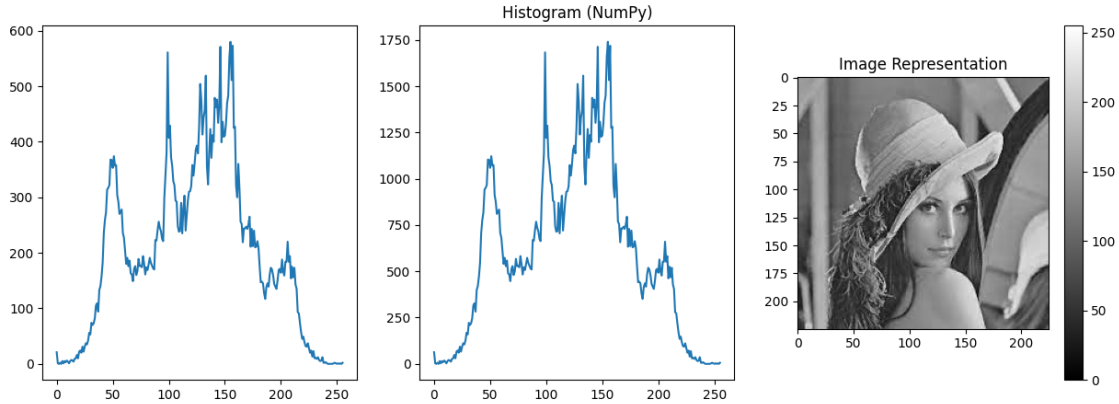
April 8, 2026

0.0.1 Take a 5x6 NumPy array and plot the histogram of the array with proper labels (without built in histogram function). Show both histograms side by side in a subplot with a title. Show input NumPy array into an image.

[18]:
```
import numpy as np

import matplotlib.pyplot as plt

array = np.array([
    [255,0,1,100,25,101],
    [10,200,70,80,120,150],
    [95,30,30,81,96,77],
    [87,89,220,250,100,10],
    [18,7,221,21,8,15]
])

hist = np.zeros(256)

for i in range(array.shape[0]):
    for j in range(array.shape[1]):
        hist[array[i,j]]+=1

hist2,bins = np.histogram(array,256,[0,256])

plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.plot(hist)

plt.subplot(1,3,2)
plt.title("Histogram (NumPy)")
plt.plot(hist2)

plt.subplot(1,3,3)
plt.title("Image Representation")
plt.imshow(array, cmap='gray')
plt.colorbar()

plt.show()
```



[19]:
```
import cv2

import numpy as np

import matplotlib.pyplot as plt

array = cv2.imread("./a.jpg")
hist = np.zeros(256)

for i in range(array.shape[0]):
    for j in range(array.shape[1]):
        hist[array[i,j]]+=1

hist2,bins = np.histogram(array,256,[0,256])

plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.plot(hist)

plt.subplot(1,3,2)
plt.title("Histogram (NumPy)")
plt.plot(hist2)

plt.subplot(1,3,3)
plt.title("Image Representation")
plt.imshow(array, cmap='gray')
plt.colorbar()
plt.show()
```



[20]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

array = cv2.imread("./a.jpg", 0)

hist = np.zeros(256)
for i in range(array.shape[0]):
    for j in range(array.shape[1]):
        hist[array[i, j]] += 1

hist2, bins = np.histogram(array.flatten(), 256, [0, 256])

equalized = cv2.equalizeHist(array)

hist_eq, bins_eq = np.histogram(equalized.flatten(), 256, [0, 256])

# Plotting
plt.figure(figsize=(18, 5))

plt.subplot(1, 4, 1)
plt.title("Original Histogram")
plt.plot(hist)

plt.subplot(1, 4, 2)
plt.title("Original Image")
plt.imshow(array, cmap='gray')

plt.subplot(1, 4, 3)
plt.title("Equalized Histogram")
plt.plot(hist_eq)
```

```
plt.subplot(1, 4, 4)
plt.title("Equalized Image")
plt.imshow(equalized, cmap='gray')

plt.tight_layout()
plt.show()
```

[21]:
```
import cv2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("./a.jpg", 0)

#  Histogram
hist = np.zeros(256)
for i in range(img.shape[0]):
    for j in range(img.shape[1]):
        hist[img[i, j]] += 1

# PDF (normalize histogram)
num_pixels = img.shape[0] * img.shape[1]
pdf = hist / num_pixels

# CDF (cumulative sum)
cdf = np.zeros(256)
cdf[0] = pdf[0]
for i in range(1, 256):
    cdf[i] = cdf[i-1] + pdf[i]

# Normalize CDF → [0,255]
cdf_normalized = np.round(cdf * 255).astype(np.uint8)

# Map old pixels → new pixels
equalized = np.zeros_like(img)
```

```python
for i in range(img.shape[0]):
    for j in range(img.shape[1]):
        equalized[i, j] = cdf_normalized[img[i, j]]

# Histograms for comparison
hist_eq, _ = np.histogram(equalized.flatten(), 256, [0, 256])


pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)

table = pd.DataFrame({
    "Old": np.arange(256),
    "PDF": pdf,
    "CDF": cdf,
    "New": cdf_normalized
})

print(table)


# Plot
plt.figure(figsize=(18,5))

plt.subplot(1,4,1)
plt.title("Original Histogram")
plt.plot(hist)

plt.subplot(1,4,2)
plt.title("Original Image")
plt.imshow(img, cmap='gray')

plt.subplot(1,4,3)
plt.title("Equalized Histogram")
plt.plot(hist_eq)

plt.subplot(1,4,4)
plt.title("Equalized Image")
plt.imshow(equalized, cmap='gray')

plt.tight_layout()
plt.show()
```

```
     Old       PDF       CDF  New
0      0  0.000415  0.000415    0
1      1  0.000020  0.000435    0
2      2  0.000000  0.000435    0
```

3	3	0.000040	0.000474	0
4	4	0.000000	0.000474	0
5	5	0.000099	0.000573	0
6	6	0.000020	0.000593	0
7	7	0.000099	0.000691	0
8	8	0.000059	0.000751	0
9	9	0.000119	0.000869	0
10	10	0.000079	0.000948	0
11	11	0.000020	0.000968	0
12	12	0.000099	0.001067	0
13	13	0.000138	0.001205	0
14	14	0.000119	0.001323	0
15	15	0.000079	0.001402	0
16	16	0.000178	0.001580	0
17	17	0.000198	0.001778	0
18	18	0.000316	0.002094	1
19	19	0.000198	0.002291	1
20	20	0.000435	0.002726	1
21	21	0.000474	0.003200	1
22	22	0.000395	0.003595	1
23	23	0.000612	0.004207	1
24	24	0.000435	0.004642	1
25	25	0.000612	0.005254	1
26	26	0.000731	0.005985	2
27	27	0.000691	0.006677	2
28	28	0.000830	0.007506	2
29	29	0.001106	0.008612	2
30	30	0.001047	0.009659	2
31	31	0.001462	0.011121	3
32	32	0.001383	0.012504	3
33	33	0.001442	0.013946	4
34	34	0.001620	0.015565	4
35	35	0.002074	0.017640	4
36	36	0.002153	0.019793	5
37	37	0.001857	0.021649	6
38	38	0.002627	0.024277	6
39	39	0.002765	0.027042	7
40	40	0.003101	0.030143	8
41	41	0.003595	0.033738	9
42	42	0.004642	0.038380	10
43	43	0.005116	0.043496	11
44	44	0.005393	0.048889	12
45	45	0.006202	0.055091	14
46	46	0.006262	0.061353	16
47	47	0.006360	0.067714	17
48	48	0.007269	0.074983	19
49	49	0.007249	0.082232	21
50	50	0.006973	0.089205	23

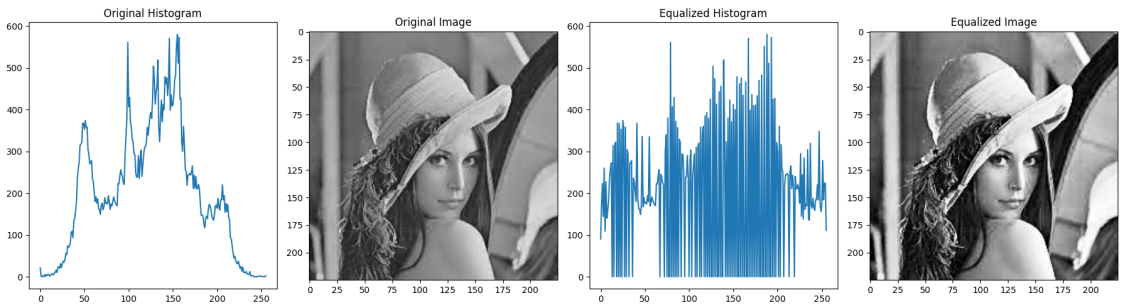
51	51	0.007388	0.096593	25
52	52	0.007052	0.103644	26
53	53	0.007072	0.110716	28
54	54	0.006005	0.116721	30
55	55	0.005768	0.122489	31
56	56	0.005333	0.127822	33
57	57	0.005412	0.133235	34
58	58	0.005491	0.138726	35
59	59	0.004662	0.143388	37
60	60	0.004484	0.147872	38
61	61	0.004030	0.151901	39
62	62	0.003556	0.155457	40
63	63	0.003773	0.159230	41
64	64	0.003477	0.162706	41
65	65	0.003674	0.166380	42
66	66	0.003220	0.169600	43
67	67	0.003180	0.172780	44
68	68	0.002943	0.175723	45
69	69	0.003378	0.179101	46
70	70	0.003477	0.182578	47
71	71	0.003160	0.185738	47
72	72	0.003299	0.189037	48
73	73	0.003733	0.192770	49
74	74	0.003477	0.196247	50
75	75	0.003496	0.199743	51
76	76	0.003437	0.203180	52
77	77	0.003832	0.207012	53
78	78	0.003556	0.210568	54
79	79	0.003180	0.213748	55
80	80	0.003437	0.217185	55
81	81	0.003338	0.220523	56
82	82	0.003536	0.224059	57
83	83	0.003773	0.227832	58
84	84	0.003595	0.231427	59
85	85	0.003496	0.234923	60
86	86	0.003398	0.238321	61
87	87	0.003358	0.241679	62
88	88	0.004405	0.246084	63
89	89	0.004385	0.250469	64
90	90	0.004760	0.255230	65
91	91	0.005057	0.260286	66
92	92	0.004840	0.265126	68
93	93	0.004721	0.269847	69
94	94	0.004444	0.274291	70
95	95	0.004365	0.278657	71
96	96	0.005906	0.284563	73
97	97	0.006716	0.291279	74
98	98	0.007506	0.298785	76

99	99	0.011081	0.309867	79
100	100	0.008040	0.317906	81
101	101	0.008474	0.326380	83
102	102	0.007348	0.333728	85
103	103	0.007052	0.340780	87
104	104	0.006499	0.347279	89
105	105	0.006360	0.353640	90
106	106	0.005807	0.359447	92
107	107	0.005768	0.365215	93
108	108	0.004899	0.370114	94
109	109	0.004701	0.374815	96
110	110	0.004721	0.379536	97
111	111	0.005728	0.385264	98
112	112	0.004642	0.389906	99
113	113	0.005511	0.395417	101
114	114	0.005985	0.401402	102
115	115	0.004741	0.406143	104
116	116	0.005294	0.411437	105
117	117	0.005807	0.417244	106
118	118	0.006123	0.423368	108
119	119	0.006143	0.429511	110
120	120	0.006321	0.435832	111
121	121	0.007072	0.442904	113
122	122	0.006696	0.449600	115
123	123	0.007111	0.456711	116
124	124	0.007605	0.464316	118
125	125	0.007763	0.472079	120
126	126	0.007486	0.479565	122
127	127	0.008395	0.487960	124
128	128	0.009956	0.497916	127
129	129	0.009363	0.507279	129
130	130	0.008158	0.515437	131
131	131	0.008751	0.524188	134
132	132	0.009007	0.533195	136
133	133	0.010252	0.543447	139
134	134	0.007032	0.550479	140
135	135	0.006380	0.556859	142
136	136	0.007506	0.564365	144
137	137	0.008356	0.572721	146
138	138	0.007328	0.580049	148
139	139	0.008138	0.588188	150
140	140	0.007901	0.596089	152
141	141	0.009462	0.605551	154
142	142	0.009126	0.614677	157
143	143	0.009402	0.624079	159
144	144	0.008573	0.632652	161
145	145	0.009205	0.641857	164
146	146	0.011279	0.653136	167

147	147	0.007881	0.661017	169
148	148	0.008612	0.669630	171
149	149	0.008079	0.677709	173
150	150	0.008119	0.685827	175
151	151	0.008553	0.694380	177
152	152	0.009284	0.703664	179
153	153	0.009541	0.713205	182
154	154	0.010904	0.724109	185
155	155	0.011457	0.735565	188
156	156	0.010094	0.745659	190
157	157	0.011319	0.756978	193
158	158	0.008395	0.765373	195
159	159	0.008435	0.773807	197
160	160	0.006341	0.780148	199
161	161	0.005926	0.786074	200
162	162	0.007111	0.793185	202
163	163	0.006242	0.799427	204
164	164	0.005057	0.804484	205
165	165	0.004978	0.809462	206
166	166	0.004326	0.813788	208
167	167	0.004820	0.818607	209
168	168	0.004820	0.823427	210
169	169	0.004879	0.828306	211
170	170	0.004800	0.833106	212
171	171	0.004958	0.838064	214
172	172	0.005235	0.843299	215
173	173	0.004168	0.847467	216
174	174	0.004800	0.852267	217
175	175	0.004188	0.856454	218
176	176	0.004741	0.861195	220
177	177	0.004148	0.865343	221
178	178	0.004168	0.869511	222
179	179	0.004365	0.873877	223
180	180	0.004128	0.878005	224
181	181	0.003477	0.881481	225
182	182	0.002904	0.884385	226
183	183	0.002923	0.887309	226
184	184	0.002864	0.890173	227
185	185	0.002509	0.892681	228
186	186	0.002311	0.894993	228
187	187	0.002726	0.897719	229
188	188	0.002864	0.900583	230
189	189	0.002746	0.903328	230
190	190	0.003180	0.906509	231
191	191	0.003417	0.909926	232
192	192	0.003358	0.913284	233
193	193	0.003160	0.916444	234
194	194	0.002844	0.919289	234

195	195	0.002746	0.922035	235
196	196	0.002667	0.924701	236
197	197	0.002963	0.927664	237
198	198	0.003338	0.931002	237
199	199	0.003378	0.934380	238
200	200	0.003240	0.937620	239
201	201	0.003714	0.941333	240
202	202	0.003279	0.944612	241
203	203	0.003141	0.947753	242
204	204	0.003635	0.951388	243
205	205	0.003635	0.955022	244
206	206	0.004346	0.959368	245
207	207	0.003635	0.963002	246
208	208	0.003832	0.966835	247
209	209	0.003042	0.969877	247
210	210	0.003457	0.973333	248
211	211	0.003081	0.976415	249
212	212	0.003417	0.979832	250
213	213	0.002825	0.982657	251
214	214	0.002667	0.985323	251
215	215	0.001837	0.987160	252
216	216	0.001798	0.988958	252
217	217	0.001442	0.990400	253
218	218	0.001126	0.991526	253
219	219	0.000889	0.992415	253
220	220	0.000968	0.993383	253
221	221	0.000770	0.994153	254
222	222	0.000632	0.994785	254
223	223	0.000612	0.995398	254
224	224	0.000731	0.996128	254
225	225	0.000553	0.996681	254
226	226	0.000395	0.997077	254
227	227	0.000474	0.997551	254
228	228	0.000257	0.997807	254
229	229	0.000454	0.998262	255
230	230	0.000198	0.998459	255
231	231	0.000217	0.998677	255
232	232	0.000178	0.998854	255
233	233	0.000237	0.999091	255
234	234	0.000138	0.999230	255
235	235	0.000099	0.999328	255
236	236	0.000119	0.999447	255
237	237	0.000237	0.999684	255
238	238	0.000040	0.999723	255
239	239	0.000040	0.999763	255
240	240	0.000059	0.999822	255
241	241	0.000020	0.999842	255
242	242	0.000000	0.999842	255

```
243   243   0.000000   0.999842   255
244   244   0.000000   0.999842   255
245   245   0.000000   0.999842   255
246   246   0.000000   0.999842   255
247   247   0.000020   0.999862   255
248   248   0.000040   0.999901   255
249   249   0.000020   0.999921   255
250   250   0.000000   0.999921   255
251   251   0.000020   0.999941   255
252   252   0.000000   0.999941   255
253   253   0.000020   0.999960   255
254   254   0.000000   0.999960   255
255   255   0.000040   1.000000   255
```



[22]:
```python
import cv2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load images (grayscale)
src = cv2.imread("./a.jpg", 0)
target = cv2.imread("./ref_image.jpg", 0)   # Target image

def compute_hist_pdf_cdf(image):
    hist = np.zeros(256)

    for i in range(image.shape[0]):
        for j in range(image.shape[1]):
            hist[image[i, j]] += 1

    num_pixels = image.shape[0] * image.shape[1]
    pdf = hist / num_pixels

    cdf = np.zeros(256)
    cdf[0] = pdf[0]
```

```
    for i in range(1, 256):
        cdf[i] = cdf[i-1] + pdf[i]

    return hist, pdf, cdf

# Source stats
hist_s, pdf_s, cdf_s = compute_hist_pdf_cdf(src)

# Target stats
hist_t, pdf_t, cdf_t = compute_hist_pdf_cdf(target)

mapping = np.zeros(256, dtype=np.uint8)

for i in range(256):
    diff = np.abs(cdf_s[i] - cdf_t)
    mapping[i] = np.argmin(diff)

matched = np.zeros_like(src)

for i in range(src.shape[0]):
    for j in range(src.shape[1]):
        matched[i, j] = mapping[src[i, j]]

# Histogram of matched image
hist_m, _, _ = compute_hist_pdf_cdf(matched)

pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)

# Source table
table_source = pd.DataFrame({
    "Gray": np.arange(256),
    "Hist_S": hist_s,
    "PDF_S": pdf_s,
    "CDF_S": cdf_s
})

# Target table
table_target = pd.DataFrame({
    "Gray": np.arange(256),
    "Hist_T": hist_t,
    "PDF_T": pdf_t,
    "CDF_T": cdf_t
})

# Mapping table
```

```python
table_mapping = pd.DataFrame({
    "Source Gray": np.arange(256),
    "Mapped Gray": mapping
})

print("\n===== SOURCE TABLE =====")
print(table_source)

print("\n===== TARGET TABLE =====")
print(table_target)

print("\n===== MAPPING TABLE =====")
print(table_mapping)

plt.figure(figsize=(18,6))

plt.subplot(2,3,1)
plt.title("Source Histogram")
plt.plot(hist_s)

plt.subplot(2,3,2)
plt.title("Target Histogram")
plt.plot(hist_t)

plt.subplot(2,3,3)
plt.title("Matched Histogram")
plt.plot(hist_m)

plt.subplot(2,3,4)
plt.title("Source Image")
plt.imshow(src, cmap='gray')

plt.subplot(2,3,5)
plt.title("Target Image")
plt.imshow(target, cmap='gray')

plt.subplot(2,3,6)
plt.title("Matched Image")
plt.imshow(matched, cmap='gray')

plt.tight_layout()
plt.show()
```

```
===== SOURCE TABLE =====
     Gray  Hist_S     PDF_S     CDF_S
0       0    21.0  0.000415  0.000415
1       1     1.0  0.000020  0.000435
```

2	2	0.0	0.000000	0.000435
3	3	2.0	0.000040	0.000474
4	4	0.0	0.000000	0.000474
5	5	5.0	0.000099	0.000573
6	6	1.0	0.000020	0.000593
7	7	5.0	0.000099	0.000691
8	8	3.0	0.000059	0.000751
9	9	6.0	0.000119	0.000869
10	10	4.0	0.000079	0.000948
11	11	1.0	0.000020	0.000968
12	12	5.0	0.000099	0.001067
13	13	7.0	0.000138	0.001205
14	14	6.0	0.000119	0.001323
15	15	4.0	0.000079	0.001402
16	16	9.0	0.000178	0.001580
17	17	10.0	0.000198	0.001778
18	18	16.0	0.000316	0.002094
19	19	10.0	0.000198	0.002291
20	20	22.0	0.000435	0.002726
21	21	24.0	0.000474	0.003200
22	22	20.0	0.000395	0.003595
23	23	31.0	0.000612	0.004207
24	24	22.0	0.000435	0.004642
25	25	31.0	0.000612	0.005254
26	26	37.0	0.000731	0.005985
27	27	35.0	0.000691	0.006677
28	28	42.0	0.000830	0.007506
29	29	56.0	0.001106	0.008612
30	30	53.0	0.001047	0.009659
31	31	74.0	0.001462	0.011121
32	32	70.0	0.001383	0.012504
33	33	73.0	0.001442	0.013946
34	34	82.0	0.001620	0.015565
35	35	105.0	0.002074	0.017640
36	36	109.0	0.002153	0.019793
37	37	94.0	0.001857	0.021649
38	38	133.0	0.002627	0.024277
39	39	140.0	0.002765	0.027042
40	40	157.0	0.003101	0.030143
41	41	182.0	0.003595	0.033738
42	42	235.0	0.004642	0.038380
43	43	259.0	0.005116	0.043496
44	44	273.0	0.005393	0.048889
45	45	314.0	0.006202	0.055091
46	46	317.0	0.006262	0.061353
47	47	322.0	0.006360	0.067714
48	48	368.0	0.007269	0.074983
49	49	367.0	0.007249	0.082232

50	50	353.0	0.006973	0.089205
51	51	374.0	0.007388	0.096593
52	52	357.0	0.007052	0.103644
53	53	358.0	0.007072	0.110716
54	54	304.0	0.006005	0.116721
55	55	292.0	0.005768	0.122489
56	56	270.0	0.005333	0.127822
57	57	274.0	0.005412	0.133235
58	58	278.0	0.005491	0.138726
59	59	236.0	0.004662	0.143388
60	60	227.0	0.004484	0.147872
61	61	204.0	0.004030	0.151901
62	62	180.0	0.003556	0.155457
63	63	191.0	0.003773	0.159230
64	64	176.0	0.003477	0.162706
65	65	186.0	0.003674	0.166380
66	66	163.0	0.003220	0.169600
67	67	161.0	0.003180	0.172780
68	68	149.0	0.002943	0.175723
69	69	171.0	0.003378	0.179101
70	70	176.0	0.003477	0.182578
71	71	160.0	0.003160	0.185738
72	72	167.0	0.003299	0.189037
73	73	189.0	0.003733	0.192770
74	74	176.0	0.003477	0.196247
75	75	177.0	0.003496	0.199743
76	76	174.0	0.003437	0.203180
77	77	194.0	0.003832	0.207012
78	78	180.0	0.003556	0.210568
79	79	161.0	0.003180	0.213748
80	80	174.0	0.003437	0.217185
81	81	169.0	0.003338	0.220523
82	82	179.0	0.003536	0.224059
83	83	191.0	0.003773	0.227832
84	84	182.0	0.003595	0.231427
85	85	177.0	0.003496	0.234923
86	86	172.0	0.003398	0.238321
87	87	170.0	0.003358	0.241679
88	88	223.0	0.004405	0.246084
89	89	222.0	0.004385	0.250469
90	90	241.0	0.004760	0.255230
91	91	256.0	0.005057	0.260286
92	92	245.0	0.004840	0.265126
93	93	239.0	0.004721	0.269847
94	94	225.0	0.004444	0.274291
95	95	221.0	0.004365	0.278657
96	96	299.0	0.005906	0.284563
97	97	340.0	0.006716	0.291279

98	98	380.0	0.007506	0.298785
99	99	561.0	0.011081	0.309867
100	100	407.0	0.008040	0.317906
101	101	429.0	0.008474	0.326380
102	102	372.0	0.007348	0.333728
103	103	357.0	0.007052	0.340780
104	104	329.0	0.006499	0.347279
105	105	322.0	0.006360	0.353640
106	106	294.0	0.005807	0.359447
107	107	292.0	0.005768	0.365215
108	108	248.0	0.004899	0.370114
109	109	238.0	0.004701	0.374815
110	110	239.0	0.004721	0.379536
111	111	290.0	0.005728	0.385264
112	112	235.0	0.004642	0.389906
113	113	279.0	0.005511	0.395417
114	114	303.0	0.005985	0.401402
115	115	240.0	0.004741	0.406143
116	116	268.0	0.005294	0.411437
117	117	294.0	0.005807	0.417244
118	118	310.0	0.006123	0.423368
119	119	311.0	0.006143	0.429511
120	120	320.0	0.006321	0.435832
121	121	358.0	0.007072	0.442904
122	122	339.0	0.006696	0.449600
123	123	360.0	0.007111	0.456711
124	124	385.0	0.007605	0.464316
125	125	393.0	0.007763	0.472079
126	126	379.0	0.007486	0.479565
127	127	425.0	0.008395	0.487960
128	128	504.0	0.009956	0.497916
129	129	474.0	0.009363	0.507279
130	130	413.0	0.008158	0.515437
131	131	443.0	0.008751	0.524188
132	132	456.0	0.009007	0.533195
133	133	519.0	0.010252	0.543447
134	134	356.0	0.007032	0.550479
135	135	323.0	0.006380	0.556859
136	136	380.0	0.007506	0.564365
137	137	423.0	0.008356	0.572721
138	138	371.0	0.007328	0.580049
139	139	412.0	0.008138	0.588188
140	140	400.0	0.007901	0.596089
141	141	479.0	0.009462	0.605551
142	142	462.0	0.009126	0.614677
143	143	476.0	0.009402	0.624079
144	144	434.0	0.008573	0.632652
145	145	466.0	0.009205	0.641857

146	146	571.0	0.011279	0.653136
147	147	399.0	0.007881	0.661017
148	148	436.0	0.008612	0.669630
149	149	409.0	0.008079	0.677709
150	150	411.0	0.008119	0.685827
151	151	433.0	0.008553	0.694380
152	152	470.0	0.009284	0.703664
153	153	483.0	0.009541	0.713205
154	154	552.0	0.010904	0.724109
155	155	580.0	0.011457	0.735565
156	156	511.0	0.010094	0.745659
157	157	573.0	0.011319	0.756978
158	158	425.0	0.008395	0.765373
159	159	427.0	0.008435	0.773807
160	160	321.0	0.006341	0.780148
161	161	300.0	0.005926	0.786074
162	162	360.0	0.007111	0.793185
163	163	316.0	0.006242	0.799427
164	164	256.0	0.005057	0.804484
165	165	252.0	0.004978	0.809462
166	166	219.0	0.004326	0.813788
167	167	244.0	0.004820	0.818607
168	168	244.0	0.004820	0.823427
169	169	247.0	0.004879	0.828306
170	170	243.0	0.004800	0.833106
171	171	251.0	0.004958	0.838064
172	172	265.0	0.005235	0.843299
173	173	211.0	0.004168	0.847467
174	174	243.0	0.004800	0.852267
175	175	212.0	0.004188	0.856454
176	176	240.0	0.004741	0.861195
177	177	210.0	0.004148	0.865343
178	178	211.0	0.004168	0.869511
179	179	221.0	0.004365	0.873877
180	180	209.0	0.004128	0.878005
181	181	176.0	0.003477	0.881481
182	182	147.0	0.002904	0.884385
183	183	148.0	0.002923	0.887309
184	184	145.0	0.002864	0.890173
185	185	127.0	0.002509	0.892681
186	186	117.0	0.002311	0.894993
187	187	138.0	0.002726	0.897719
188	188	145.0	0.002864	0.900583
189	189	139.0	0.002746	0.903328
190	190	161.0	0.003180	0.906509
191	191	173.0	0.003417	0.909926
192	192	170.0	0.003358	0.913284
193	193	160.0	0.003160	0.916444

194	194	144.0	0.002844	0.919289
195	195	139.0	0.002746	0.922035
196	196	135.0	0.002667	0.924701
197	197	150.0	0.002963	0.927664
198	198	169.0	0.003338	0.931002
199	199	171.0	0.003378	0.934380
200	200	164.0	0.003240	0.937620
201	201	188.0	0.003714	0.941333
202	202	166.0	0.003279	0.944612
203	203	159.0	0.003141	0.947753
204	204	184.0	0.003635	0.951388
205	205	184.0	0.003635	0.955022
206	206	220.0	0.004346	0.959368
207	207	184.0	0.003635	0.963002
208	208	194.0	0.003832	0.966835
209	209	154.0	0.003042	0.969877
210	210	175.0	0.003457	0.973333
211	211	156.0	0.003081	0.976415
212	212	173.0	0.003417	0.979832
213	213	143.0	0.002825	0.982657
214	214	135.0	0.002667	0.985323
215	215	93.0	0.001837	0.987160
216	216	91.0	0.001798	0.988958
217	217	73.0	0.001442	0.990400
218	218	57.0	0.001126	0.991526
219	219	45.0	0.000889	0.992415
220	220	49.0	0.000968	0.993383
221	221	39.0	0.000770	0.994153
222	222	32.0	0.000632	0.994785
223	223	31.0	0.000612	0.995398
224	224	37.0	0.000731	0.996128
225	225	28.0	0.000553	0.996681
226	226	20.0	0.000395	0.997077
227	227	24.0	0.000474	0.997551
228	228	13.0	0.000257	0.997807
229	229	23.0	0.000454	0.998262
230	230	10.0	0.000198	0.998459
231	231	11.0	0.000217	0.998677
232	232	9.0	0.000178	0.998854
233	233	12.0	0.000237	0.999091
234	234	7.0	0.000138	0.999230
235	235	5.0	0.000099	0.999328
236	236	6.0	0.000119	0.999447
237	237	12.0	0.000237	0.999684
238	238	2.0	0.000040	0.999723
239	239	2.0	0.000040	0.999763
240	240	3.0	0.000059	0.999822
241	241	1.0	0.000020	0.999842

242	242	0.0	0.000000	0.999842
243	243	0.0	0.000000	0.999842
244	244	0.0	0.000000	0.999842
245	245	0.0	0.000000	0.999842
246	246	0.0	0.000000	0.999842
247	247	1.0	0.000020	0.999862
248	248	2.0	0.000040	0.999901
249	249	1.0	0.000020	0.999921
250	250	0.0	0.000000	0.999921
251	251	1.0	0.000020	0.999941
252	252	0.0	0.000000	0.999941
253	253	1.0	0.000020	0.999960
254	254	0.0	0.000000	0.999960
255	255	2.0	0.000040	1.000000

===== TARGET TABLE =====

	Gray	Hist_T	PDF_T	CDF_T
0	0	413.0	0.008158	0.008158
1	1	70.0	0.001383	0.009541
2	2	95.0	0.001877	0.011417
3	3	105.0	0.002074	0.013491
4	4	112.0	0.002212	0.015704
5	5	137.0	0.002706	0.018410
6	6	144.0	0.002844	0.021254
7	7	202.0	0.003990	0.025244
8	8	312.0	0.006163	0.031407
9	9	444.0	0.008770	0.040178
10	10	609.0	0.012030	0.052207
11	11	764.0	0.015091	0.067299
12	12	1079.0	0.021314	0.088612
13	13	1411.0	0.027872	0.116484
14	14	1395.0	0.027556	0.144040
15	15	581.0	0.011477	0.155516
16	16	514.0	0.010153	0.165669
17	17	297.0	0.005867	0.171536
18	18	222.0	0.004385	0.175921
19	19	162.0	0.003200	0.179121
20	20	150.0	0.002963	0.182084
21	21	118.0	0.002331	0.184415
22	22	123.0	0.002430	0.186844
23	23	104.0	0.002054	0.188899
24	24	79.0	0.001560	0.190459
25	25	94.0	0.001857	0.192316
26	26	88.0	0.001738	0.194054
27	27	80.0	0.001580	0.195635
28	28	90.0	0.001778	0.197412
29	29	71.0	0.001402	0.198815
30	30	75.0	0.001481	0.200296

31	31	73.0	0.001442	0.201738
32	32	79.0	0.001560	0.203299
33	33	81.0	0.001600	0.204899
34	34	57.0	0.001126	0.206025
35	35	68.0	0.001343	0.207368
36	36	72.0	0.001422	0.208790
37	37	62.0	0.001225	0.210015
38	38	60.0	0.001185	0.211200
39	39	52.0	0.001027	0.212227
40	40	60.0	0.001185	0.213412
41	41	70.0	0.001383	0.214795
42	42	68.0	0.001343	0.216138
43	43	70.0	0.001383	0.217521
44	44	52.0	0.001027	0.218548
45	45	60.0	0.001185	0.219733
46	46	48.0	0.000948	0.220681
47	47	58.0	0.001146	0.221827
48	48	57.0	0.001126	0.222953
49	49	63.0	0.001244	0.224198
50	50	52.0	0.001027	0.225225
51	51	54.0	0.001067	0.226291
52	52	49.0	0.000968	0.227259
53	53	53.0	0.001047	0.228306
54	54	58.0	0.001146	0.229452
55	55	59.0	0.001165	0.230617
56	56	61.0	0.001205	0.231822
57	57	55.0	0.001086	0.232909
58	58	60.0	0.001185	0.234094
59	59	56.0	0.001106	0.235200
60	60	43.0	0.000849	0.236049
61	61	61.0	0.001205	0.237254
62	62	56.0	0.001106	0.238360
63	63	53.0	0.001047	0.239407
64	64	45.0	0.000889	0.240296
65	65	41.0	0.000810	0.241106
66	66	57.0	0.001126	0.242232
67	67	53.0	0.001047	0.243279
68	68	45.0	0.000889	0.244168
69	69	49.0	0.000968	0.245136
70	70	47.0	0.000928	0.246064
71	71	43.0	0.000849	0.246914
72	72	51.0	0.001007	0.247921
73	73	45.0	0.000889	0.248810
74	74	37.0	0.000731	0.249541
75	75	40.0	0.000790	0.250331
76	76	50.0	0.000988	0.251319
77	77	34.0	0.000672	0.251990
78	78	38.0	0.000751	0.252741

79	79	44.0	0.000869	0.253610
80	80	43.0	0.000849	0.254459
81	81	44.0	0.000869	0.255328
82	82	35.0	0.000691	0.256020
83	83	50.0	0.000988	0.257007
84	84	42.0	0.000830	0.257837
85	85	50.0	0.000988	0.258825
86	86	43.0	0.000849	0.259674
87	87	53.0	0.001047	0.260721
88	88	47.0	0.000928	0.261649
89	89	60.0	0.001185	0.262835
90	90	62.0	0.001225	0.264059
91	91	67.0	0.001323	0.265383
92	92	67.0	0.001323	0.266706
93	93	63.0	0.001244	0.267951
94	94	89.0	0.001758	0.269709
95	95	87.0	0.001719	0.271427
96	96	84.0	0.001659	0.273086
97	97	81.0	0.001600	0.274686
98	98	98.0	0.001936	0.276622
99	99	128.0	0.002528	0.279151
100	100	124.0	0.002449	0.281600
101	101	120.0	0.002370	0.283970
102	102	126.0	0.002489	0.286459
103	103	148.0	0.002923	0.289383
104	104	144.0	0.002844	0.292227
105	105	192.0	0.003793	0.296020
106	106	173.0	0.003417	0.299437
107	107	157.0	0.003101	0.302538
108	108	181.0	0.003575	0.306114
109	109	180.0	0.003556	0.309669
110	110	198.0	0.003911	0.313580
111	111	215.0	0.004247	0.317827
112	112	209.0	0.004128	0.321956
113	113	237.0	0.004681	0.326637
114	114	236.0	0.004662	0.331299
115	115	255.0	0.005037	0.336336
116	116	294.0	0.005807	0.342143
117	117	288.0	0.005689	0.347832
118	118	269.0	0.005314	0.353146
119	119	303.0	0.005985	0.359131
120	120	265.0	0.005235	0.364365
121	121	299.0	0.005906	0.370272
122	122	269.0	0.005314	0.375585
123	123	274.0	0.005412	0.380998
124	124	302.0	0.005965	0.386963
125	125	330.0	0.006519	0.393481
126	126	336.0	0.006637	0.400119

127	127	315.0	0.006222	0.406341
128	128	377.0	0.007447	0.413788
129	129	346.0	0.006835	0.420622
130	130	330.0	0.006519	0.427141
131	131	338.0	0.006677	0.433817
132	132	330.0	0.006519	0.440336
133	133	359.0	0.007091	0.447427
134	134	344.0	0.006795	0.454222
135	135	309.0	0.006104	0.460326
136	136	280.0	0.005531	0.465857
137	137	307.0	0.006064	0.471921
138	138	276.0	0.005452	0.477373
139	139	259.0	0.005116	0.482489
140	140	269.0	0.005314	0.487802
141	141	249.0	0.004919	0.492721
142	142	286.0	0.005649	0.498370
143	143	296.0	0.005847	0.504217
144	144	255.0	0.005037	0.509254
145	145	331.0	0.006538	0.515793
146	146	330.0	0.006519	0.522311
147	147	421.0	0.008316	0.530627
148	148	464.0	0.009165	0.539793
149	149	415.0	0.008198	0.547990
150	150	486.0	0.009600	0.557590
151	151	363.0	0.007170	0.564760
152	152	386.0	0.007625	0.572385
153	153	407.0	0.008040	0.580425
154	154	442.0	0.008731	0.589156
155	155	435.0	0.008593	0.597748
156	156	580.0	0.011457	0.609205
157	157	543.0	0.010726	0.619931
158	158	580.0	0.011457	0.631388
159	159	577.0	0.011398	0.642785
160	160	544.0	0.010746	0.653531
161	161	735.0	0.014519	0.668049
162	162	827.0	0.016336	0.684385
163	163	1252.0	0.024731	0.709116
164	164	908.0	0.017936	0.727052
165	165	832.0	0.016435	0.743486
166	166	791.0	0.015625	0.759111
167	167	919.0	0.018153	0.777264
168	168	657.0	0.012978	0.790242
169	169	772.0	0.015249	0.805491
170	170	423.0	0.008356	0.813847
171	171	519.0	0.010252	0.824099
172	172	531.0	0.010489	0.834588
173	173	560.0	0.011062	0.845649
174	174	577.0	0.011398	0.857047

175	175	515.0	0.010173	0.867220
176	176	509.0	0.010054	0.877274
177	177	442.0	0.008731	0.886005
178	178	411.0	0.008119	0.894123
179	179	503.0	0.009936	0.904059
180	180	602.0	0.011891	0.915951
181	181	520.0	0.010272	0.926222
182	182	593.0	0.011714	0.937936
183	183	523.0	0.010331	0.948267
184	184	508.0	0.010035	0.958301
185	185	417.0	0.008237	0.966538
186	186	262.0	0.005175	0.971714
187	187	196.0	0.003872	0.975585
188	188	157.0	0.003101	0.978686
189	189	112.0	0.002212	0.980899
190	190	52.0	0.001027	0.981926
191	191	57.0	0.001126	0.983052
192	192	61.0	0.001205	0.984257
193	193	45.0	0.000889	0.985146
194	194	46.0	0.000909	0.986054
195	195	32.0	0.000632	0.986686
196	196	38.0	0.000751	0.987437
197	197	28.0	0.000553	0.987990
198	198	23.0	0.000454	0.988444
199	199	28.0	0.000553	0.988998
200	200	19.0	0.000375	0.989373
201	201	27.0	0.000533	0.989906
202	202	28.0	0.000553	0.990459
203	203	30.0	0.000593	0.991052
204	204	15.0	0.000296	0.991348
205	205	21.0	0.000415	0.991763
206	206	19.0	0.000375	0.992138
207	207	15.0	0.000296	0.992435
208	208	15.0	0.000296	0.992731
209	209	17.0	0.000336	0.993067
210	210	9.0	0.000178	0.993244
211	211	11.0	0.000217	0.993462
212	212	18.0	0.000356	0.993817
213	213	11.0	0.000217	0.994035
214	214	12.0	0.000237	0.994272
215	215	7.0	0.000138	0.994410
216	216	8.0	0.000158	0.994568
217	217	10.0	0.000198	0.994765
218	218	4.0	0.000079	0.994844
219	219	5.0	0.000099	0.994943
220	220	7.0	0.000138	0.995081
221	221	9.0	0.000178	0.995259
222	222	4.0	0.000079	0.995338

223	223	10.0	0.000198	0.995536
224	224	2.0	0.000040	0.995575
225	225	5.0	0.000099	0.995674
226	226	14.0	0.000277	0.995951
227	227	10.0	0.000198	0.996148
228	228	6.0	0.000119	0.996267
229	229	7.0	0.000138	0.996405
230	230	9.0	0.000178	0.996583
231	231	9.0	0.000178	0.996760
232	232	10.0	0.000198	0.996958
233	233	7.0	0.000138	0.997096
234	234	9.0	0.000178	0.997274
235	235	11.0	0.000217	0.997491
236	236	8.0	0.000158	0.997649
237	237	9.0	0.000178	0.997827
238	238	11.0	0.000217	0.998044
239	239	11.0	0.000217	0.998262
240	240	7.0	0.000138	0.998400
241	241	4.0	0.000079	0.998479
242	242	12.0	0.000237	0.998716
243	243	4.0	0.000079	0.998795
244	244	6.0	0.000119	0.998914
245	245	6.0	0.000119	0.999032
246	246	8.0	0.000158	0.999190
247	247	7.0	0.000138	0.999328
248	248	4.0	0.000079	0.999407
249	249	6.0	0.000119	0.999526
250	250	4.0	0.000079	0.999605
251	251	1.0	0.000020	0.999625
252	252	3.0	0.000059	0.999684
253	253	3.0	0.000059	0.999743
254	254	1.0	0.000020	0.999763
255	255	12.0	0.000237	1.000000

===== MAPPING TABLE =====

	Source Gray	Mapped Gray
0	0	0
1	1	0
2	2	0
3	3	0
4	4	0
5	5	0
6	6	0
7	7	0
8	8	0
9	9	0
10	10	0
11	11	0

12	12	0
13	13	0
14	14	0
15	15	0
16	16	0
17	17	0
18	18	0
19	19	0
20	20	0
21	21	0
22	22	0
23	23	0
24	24	0
25	25	0
26	26	0
27	27	0
28	28	0
29	29	0
30	30	1
31	31	2
32	32	3
33	33	3
34	34	4
35	35	5
36	36	5
37	37	6
38	38	7
39	39	7
40	40	8
41	41	8
42	42	9
43	43	9
44	44	10
45	45	10
46	46	11
47	47	11
48	48	11
49	49	12
50	50	12
51	51	12
52	52	13
53	53	13
54	54	13
55	55	13
56	56	13
57	57	14
58	58	14
59	59	14

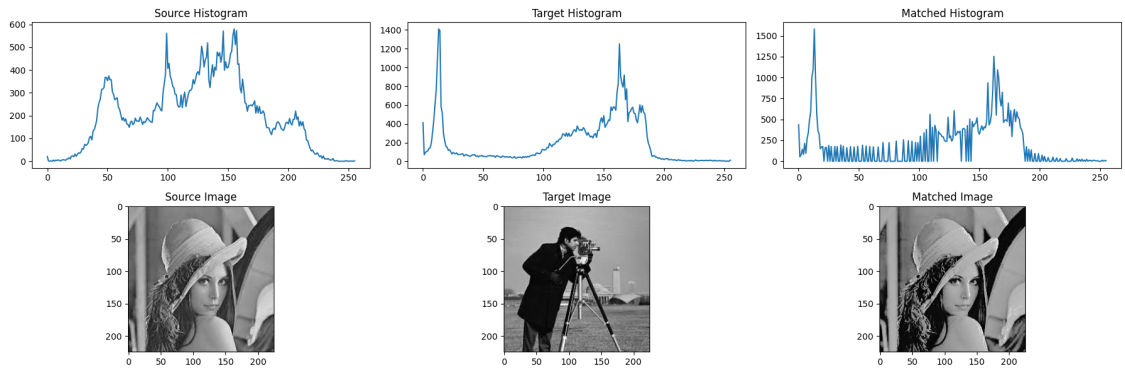
60	60	14
61	61	15
62	62	15
63	63	15
64	64	16
65	65	16
66	66	17
67	67	17
68	68	18
69	69	19
70	70	20
71	71	22
72	72	23
73	73	25
74	74	27
75	75	30
76	76	32
77	77	35
78	78	37
79	79	40
80	80	43
81	81	46
82	82	49
83	83	53
84	84	56
85	85	59
86	86	62
87	87	66
88	88	70
89	89	75
90	90	81
91	91	87
92	92	91
93	93	94
94	94	97
95	95	99
96	96	101
97	97	104
98	98	106
99	99	109
100	100	111
101	101	113
102	102	114
103	103	116
104	104	117
105	105	118
106	106	119
107	107	120

108	108	121
109	109	122
110	110	123
111	111	124
112	112	124
113	113	125
114	114	126
115	115	127
116	116	128
117	117	129
118	118	129
119	119	130
120	120	131
121	121	132
122	122	133
123	123	134
124	124	136
125	125	137
126	126	138
127	127	140
128	128	142
129	129	144
130	130	145
131	131	146
132	132	147
133	133	148
134	134	149
135	135	150
136	136	151
137	137	152
138	138	153
139	139	154
140	140	155
141	141	156
142	142	157
143	143	157
144	144	158
145	145	159
146	146	160
147	147	161
148	148	161
149	149	162
150	150	162
151	151	162
152	152	163
153	153	163
154	154	164
155	155	165

156	156	165
157	157	166
158	158	166
159	159	167
160	160	167
161	161	168
162	162	168
163	163	169
164	164	169
165	165	169
166	166	170
167	167	170
168	168	171
169	169	171
170	170	172
171	171	172
172	172	173
173	173	173
174	174	174
175	175	174
176	176	174
177	177	175
178	178	175
179	179	176
180	180	176
181	181	176
182	182	177
183	183	177
184	184	178
185	185	178
186	186	178
187	187	178
188	188	179
189	189	179
190	190	179
191	191	179
192	192	180
193	193	180
194	194	180
195	195	181
196	196	181
197	197	181
198	198	181
199	199	182
200	200	182
201	201	182
202	202	183
203	203	183

204	204	183
205	205	184
206	206	184
207	207	185
208	208	185
209	209	186
210	210	186
211	211	187
212	212	189
213	213	191
214	214	193
215	215	196
216	216	199
217	217	202
218	218	204
219	219	207
220	220	211
221	221	214
222	222	217
223	223	222
224	224	227
225	225	231
226	226	233
227	227	235
228	228	237
229	229	239
230	230	241
231	231	242
232	232	244
233	233	245
234	234	246
235	235	247
236	236	248
237	237	252
238	238	253
239	239	254
240	240	254
241	241	254
242	242	254
243	243	254
244	244	254
245	245	254
246	246	254
247	247	254
248	248	255
249	249	255
250	250	255
251	251	255

252	252	255
253	253	255
254	254	255
255	255	255



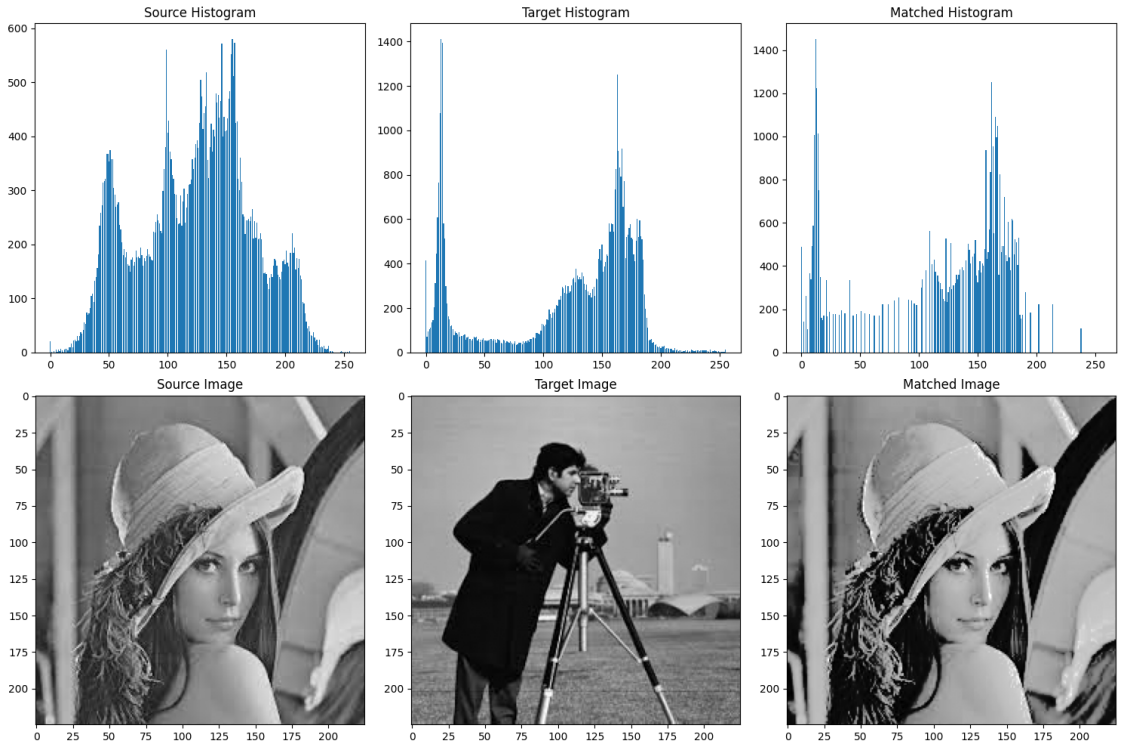
[30]:
```python
import cv2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load images (grayscale)
src = cv2.imread("./a.jpg", 0)
target = cv2.imread("./ref_image.jpg", 0)

def manual_histogram_processing(image):
    #  Compute Histogram (n_k)
    hist = np.zeros(256)
    for i in range(image.shape[0]):
        for j in range(image.shape[1]):
            hist[image[i, j]] += 1

    #  Compute PDF (p_r = n_k / MN)
    num_pixels = image.shape[0] * image.shape[1]
    pdf = hist / num_pixels

    #  Compute CDF (Sum of PDF)
    cdf = np.cumsum(pdf)

    # Histogram Equalization Scaling (s = (L-1) * CDF)
    # We round to the nearest integer as per manual calculation rules in the PDF
    equalized_values = np.round(cdf * 255).astype(np.uint8)

    return hist, pdf, cdf, equalized_values
```

```python
# --- Process Source Image ---
hist_s, pdf_s, cdf_s, s_k = manual_histogram_processing(src)

# --- Process Target Image ---
hist_t, pdf_t, cdf_t, v_q = manual_histogram_processing(target)

# --- Mapping (Inverse Transformation) ---
# We find z such that G(z) is closest to T(r)
mapping = np.zeros(256, dtype=np.uint8)

for i in range(256):
    # Casting to float prevents the unsigned integer wrap-around
    s_val = float(s_k[i])
    min_diff = float('inf')
    best_index = 0

    for q in range(256):
        v_val = float(v_q[q])
        current_diff = abs(s_val - v_val)

        if current_diff < min_diff:
            min_diff = current_diff
            best_index = q

    mapping[i] = best_index

# --- Apply Mapping to Create Matched Image ---
matched = mapping[src]

# Calculate final histogram for verification
hist_m, _, _, _ = manual_histogram_processing(matched)


table_source = pd.DataFrame({
    "Gray_r": np.arange(256),
    "Count_nk": hist_s,
    "PDF_Pr": pdf_s,
    "CDF": cdf_s,
    "Eq_sk": s_k
})

table_target = pd.DataFrame({
    "Gray_z": np.arange(256),
    "Count_nk": hist_t,
    "PDF_Pz": pdf_t,
    "CDF": cdf_t,
    "Eq_vq": v_q
```

```python
})

mapping_summary = pd.DataFrame({
    "Source_Gray": np.arange(256),
    "Mapped_to_Target": mapping
})

print("\n===== SOURCE CALCULATION =====")
print(table_source)

print("\n===== TARGET CALCULATION =====")
print(table_target)

print("\n===== FINAL MAPPING =====")
print(mapping_summary)

# --- VISUALIZATION ---
plt.figure(figsize=(15, 10))

plt.subplot(2, 3, 1); plt.title("Source Histogram"); plt.bar(range(256), hist_s)
plt.subplot(2, 3, 2); plt.title("Target Histogram"); plt.bar(range(256), hist_t)
plt.subplot(2, 3, 3); plt.title("Matched Histogram"); plt.bar(range(256),␣
 ↪hist_m)

plt.subplot(2, 3, 4); plt.title("Source Image"); plt.imshow(src, cmap='gray')
plt.subplot(2, 3, 5); plt.title("Target Image"); plt.imshow(target, cmap='gray')
plt.subplot(2, 3, 6); plt.title("Matched Image"); plt.imshow(matched,␣
 ↪cmap='gray')

plt.tight_layout()
plt.show()
```

```
===== SOURCE CALCULATION =====
     Gray_r  Count_nk    PDF_Pr       CDF  Eq_sk
0         0      21.0  0.000415  0.000415      0
1         1       1.0  0.000020  0.000435      0
2         2       0.0  0.000000  0.000435      0
3         3       2.0  0.000040  0.000474      0
4         4       0.0  0.000000  0.000474      0
..      …       …        …        …    …
251     251       1.0  0.000020  0.999941    255
252     252       0.0  0.000000  0.999941    255
253     253       1.0  0.000020  0.999960    255
254     254       0.0  0.000000  0.999960    255
255     255       2.0  0.000040  1.000000    255
```

```
[256 rows x 5 columns]

===== TARGET CALCULATION =====
     Gray_z  Count_nk    PDF_Pz       CDF  Eq_vq
0         0     413.0  0.008158  0.008158      2
1         1      70.0  0.001383  0.009541      2
2         2      95.0  0.001877  0.011417      3
3         3     105.0  0.002074  0.013491      3
4         4     112.0  0.002212  0.015704      4
..      …       …       …       …    …
251     251       1.0  0.000020  0.999625    255
252     252       3.0  0.000059  0.999684    255
253     253       3.0  0.000059  0.999743    255
254     254       1.0  0.000020  0.999763    255
255     255      12.0  0.000237  1.000000    255

[256 rows x 5 columns]

===== FINAL MAPPING =====
     Source_Gray  Mapped_to_Target
0              0                 0
1              1                 0
2              2                 0
3              3                 0
4              4                 0
..           …               …
251          251               238
252          252               238
253          253               238
254          254               238
255          255               238

[256 rows x 2 columns]
```


Source Histogram
Target Histogram
Matched Histogram
Source Image
Target Image
Matched Image